

SONAKSHI CHOWGULE- CNL ASSIGNMENT 03A

```
import java.util.Scanner;

public class IPAddressInfo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("Subnet Mask : A 32-bit value used to identify the network and host
portions of an IP address.");
        System.out.println("Subnetting : The process of dividing one large network into multiple
smaller, manageable subnetworks.");
        System.out.println("-----");
        System.out.println("IP Address : A unique logical address assigned to a device on a
network.");
        System.out.println("IP Class : A category of IP addresses based on the first octet.");
        System.out.println("Subnet Mask : A 32-bit number that separates the Network ID and Host
ID.");
        System.out.println("Network ID : The portion of an IP address that identifies the network.");
        System.out.println("Host ID : The portion of an IP address that identifies a device within a
network.");
        System.out.println("Subnetting : The process of dividing a network into smaller
subnetworks.");
        System.out.println("Network Bits: Bits used to identify the network portion of an IP
address.");
        System.out.println("Host Bits : Bits used to identify hosts within a network.");
        System.out.println("Hosts/Network: Maximum usable hosts in a network (2^HostBits - 2).");
        System.out.println("No. of Networks: Total possible networks in a class
(2^AvailableNetworkBits).");
        System.out.println("Default Subnet Mask: The predefined subnet mask assigned to an IP
class.");
        System.out.println("IP Range : The minimum and maximum IP addresses belonging to a
class.");
        System.out.println("Octet : Each 8-bit section of an IPv4 address.");
        System.out.println("-----");

        System.out.print("Enter an IPv4 Address: ");
        String ip = sc.nextLine();

        String[] parts = ip.split("\\.");

        if (parts.length != 4) {
            System.out.println("Invalid IP Address!");
        }
    }
}
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        return;
    }

    int firstOctet;

    try {
        firstOctet = Integer.parseInt(parts[0]);

        for (String part : parts) {
            int num = Integer.parseInt(part);
            if (num < 0 || num > 255) {
                System.out.println("Invalid IP Address!");
                return;
            }
        }
    } catch (NumberFormatException e) {
        System.out.println("Invalid IP Address!");
        return;
    }

    if (firstOctet >= 1 && firstOctet <= 126) {
        System.out.println("Class : A");
        System.out.println("Decimal Range : 0 - 127");
        System.out.println("1st Octet Range : 1 - 126");
        System.out.println("IP Address Range : 1.0.0.0 - 126.255.255.255");
        System.out.println("Subnet Mask : 255.0.0.0");
        System.out.println("Hosts/Network ID : 2^24 - 2");
        System.out.println("No. of Networks : 2^7 (128)");
    }
    else if (firstOctet >= 128 && firstOctet <= 191) {
        System.out.println("Class : B");
        System.out.println("Decimal Range : 128 - 191");
        System.out.println("1st Octet Range : 128 - 191");
        System.out.println("IP Address Range : 128.0.0.0 - 191.255.255.255");
        System.out.println("Subnet Mask : 255.255.0.0");
        System.out.println("Hosts/Network ID : 2^16 - 2");
        System.out.println("No. of Networks : 2^14 (16384)");
    }
    else if (firstOctet >= 192 && firstOctet <= 223) {
        System.out.println("Class : C");
        System.out.println("Decimal Range : 192 - 223");
        System.out.println("1st Octet Range : 192 - 223");
        System.out.println("IP Address Range : 192.0.0.0 - 223.255.255.255");
        System.out.println("Subnet Mask : 255.255.255.0");
    }

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        System.out.println("Hosts/Network ID : 2^8 - 2");
        System.out.println("No. of Networks : 2^21 (2097152)");
    }
    else if (firstOctet >= 224 && firstOctet <= 239) {
        System.out.println("Class : D");
        System.out.println("Decimal Range : 224 - 239");
        System.out.println("1st Octet Range : 224 - 239");
        System.out.println("IP Address Range : 224.0.0.0 - 239.255.255.255");
        System.out.println("Subnet Mask : N/A");
        System.out.println("Hosts/Network ID : Multicast");
        System.out.println("No. of Networks : N/A");
    }
    else if (firstOctet >= 240 && firstOctet <= 255) {
        System.out.println("Class : E");
        System.out.println("Decimal Range : 240 - 255");
        System.out.println("1st Octet Range : 240 - 255");
        System.out.println("IP Address Range : 240.0.0.0 - 255.255.255.255");
        System.out.println("Subnet Mask : N/A");
        System.out.println("Hosts/Network ID : Experimental");
        System.out.println("No. of Networks : N/A");
    }else {
        System.out.println("Invalid IP Address!");
    }

    sc.close();
}
}

```

OUTPUT:

Subnet Mask : A 32-bit value used to identify the network and host portions of an IP address.

Subnetting : The process of dividing one large network into multiple smaller, manageable subnetworks.

IP Address : A unique logical address assigned to a device on a network.

IP Class : A category of IP addresses based on the first octet.

Subnet Mask : A 32-bit number that separates the Network ID and Host ID.

Network ID : The portion of an IP address that identifies the network.

Host ID : The portion of an IP address that identifies a device within a network.

Subnetting : The process of dividing a network into smaller subnetworks.

Network Bits: Bits used to identify the network portion of an IP address.

Host Bits : Bits used to identify hosts within a network.

Hosts/Network: Maximum usable hosts in a network ($2^{\text{HostBits}} - 2$).

No. of Networks: Total possible networks in a class ($2^{\text{AvailableNetworkBits}}$).

Default Subnet Mask: The predefined subnet mask assigned to an IP class.

IP Range : The minimum and maximum IP addresses belonging to a class.

Octet : Each 8-bit section of an IPv4 address.

Enter an IPv4 Address: 192.168.1.10

Class : C

Decimal Range : 192 - 223

1st Octet Range : 192 - 223

IP Address Range : 192.0.0.0 - 223.255.255.255

Subnet Mask : 255.255.255.0

Hosts/Network ID : $2^8 - 2$

No. of Networks : 2^{21} (2097152)

GITHUB LINK:

https://github.com/chowgulesonakshi-lab/Computer-Networks/blob/main/Subnetting_A